

MTH212M PRACTICE PROBLEMS WEEK 6

Question 1. Work out the exercises left out in class.

Question 2. Recall that the simple symmetric Random walk is recurrent in one dimension. Check if it is null/positive recurrent.

Question 3. Consider a Markov chain on the state space $\{0, 1, 2, \dots\}$ and the T.P. matrix

$$\begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 & \cdots \\ a_0 & a_1 & a_2 & a_3 & a_4 & \cdots \\ 0 & a_0 & a_1 & a_2 & a_3 & \cdots \\ 0 & 0 & a_0 & a_1 & a_2 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \cdots \end{pmatrix}$$

with all $a_n > 0$ and $\sum_{n=0}^{\infty} a_n = 1$. Show that the chain is transient if and only if $\sum_{n=0}^{\infty} na_n > 1$. (equivalently, the chain is recurrent if and only if $\sum_{n=0}^{\infty} na_n \leq 1$.)

Question 4. Fix $p \in (0, 1)$ and set $q := 1 - p$. Consider a Markov Chain with the state space $\{0, 1, \dots\}$ and transition probability matrix

$$P = \begin{pmatrix} p & q & 0 & 0 & 0 & \cdots \\ p & 0 & q & 0 & 0 & \cdots \\ 0 & p & 0 & q & 0 & \cdots \\ 0 & 0 & p & 0 & q & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Check that the MC is irreducible and aperiodic. Is it positive recurrent?